AVISHKAAR MAKEATHON 2026 · HEALTH GUARD

# NeuroRide

An eye- and brain-controlled smart wheelchair with health monitoring and automatic medical emergency response.

Theme: Mayday — The Final Call

**Bhuvan Gowda BC**
Bengaluru · October 2026

## 1 Project summary

> NeuroRide lets a person with severe paralysis drive a wheelchair using only eye movements and blinks, while it watches their health and calls for help.

- **Problem:** people with spinal injuries, ALS or stroke often cannot use a joystick. In a flood or emergency they are the most likely to be stranded.
- **Solution:** a BioAmp EXG Pill reads eye signals (EOG) and jaw clench (EMG). An ESP32 turns them into drive commands, with an obstacle stop that overrides everything.
- **Health Guard:** heart rate and SpO2, fall/tilt detection, automatic SOS with location.
- **Mayday theme fit:** a water sensor under the chair locks movement into deep water and alerts rescuers.
- **Prototype form:** a scale model built on the Avishkaar Pro kit chassis.

## 2 Kit check: DIY Neuroscience Kit – Basic

The kit covers all the biosignal parts, but not the controller, drive electronics or health sensors. Price as of 10 Oct 2026: ₹5,999 without board, ₹6,499 with Arduino UNO R4 Minima.

| Item in the box | Used for in NeuroRide |
|---|---|
| BioAmp EXG Pill | Main amplifier for EOG/EMG drive control |
| 2-ch Brain BioAmp Band | Electrode band for EEG / forehead EOG |
| Muscle BioAmp Band | Jaw-clench EMG (emergency stop) |
| Heart BioAmp Band | Optional ECG demo |
| BioAmp cable (100 cm) | Electrodes to pill |
| Gel electrodes (24 + 100 boxy + 3 repositionable) | Skin contact |
| Electrode gel, NuPrep skin prep gel, wipes | Good signal quality |
| 3 jumper cables | Pill to board |

**Not in the kit:** ESP32-WROOM-32 board, jumper wire packs, breadboard, USB data cable, motor driver, MAX30102, MPU6050, HC-SR04, water sensor, buzzer, batteries.

**Compatibility:** Upside Down Labs publishes ESP32 wiring and filter sketches for EOG, EMG and EEG, plus an EOG blink-detection example. Power the pill at 3.3 V on the ESP32 so the ADC pin never sees more than 3.3 V.

## 3 Prior art and what makes NeuroRide different

> The core blink-control idea already exists, so judges will score us on what we add.

| Existing work | What it does |
|---|---|
| Smart Wheel Chair, Avishkaar community (Shiv Soni, Mar 2024) | Brain-controlled wheelchair on motors and wheels for people who cannot walk |
| Smart Assistive System for Paralysis People (IJRASET) | BioAmp EXG Pill + ESP32; one blink moves forward, no blink stops; web control for caregivers |
| Eye Controlled Wheel Chair Using EOG | EOG eye movements switch relays to steer |

### Our differentiators

1. Multi-input control: glance to steer, double blink to start/stop, jaw clench to stop, plus app and voice backup.
2. Safety layer: ultrasonic obstacle stop and tilt cut-off override every brain command.
3. Health Guard: live heart rate, SpO2, fall detection and auto-SOS with location.
4. Medical Emergency Response: auto-alerts, doctor report and first-aid guidance.
5. Flood mode: water sensor locks movement and alerts rescuers.
6. Per-user calibration: a 30-second setup that learns each person's signals, reducing false triggers — the open problem the IJRASET paper names.

## 4 Medical Emergency Response

NeuroRide cuts the time between an emergency and help arriving: it detects the problem, calls for help, sends the doctor a report, and guides people nearby until help comes.

### 1 · Detect automatically

- Heart rate or SpO2 outside safe limits for 30+ seconds
- Fall or tilt detected (MPU6050)
- No response to "Are you OK? Blink twice" within 15 s
- Manual SOS by a blink or jaw-clench pattern

### 2 · Call for help

- SMS + call to family and caregiver with live GPS
- One-tap 108 ambulance call with details ready
- Buzzer and flashing light on the chair
- Optional SIM800L GSM module (~₹400)

### 3 · Auto doctor report

- Last 30 min of heart rate and SpO2, event time, fall data, location
- Medical ID: blood group, conditions, medicines, allergies
- Sent as a link before the patient reaches hospital

### 4 · Guidance until help arrives

- Step-by-step first aid by voice and on screen
- One-tap video call to an online doctor
- Ambulance arrival countdown

**Safety notes:** first-aid guidance supports trained help and never replaces it; it should be reviewed by a doctor before real use. In the demo all emergency calls are simulated — never dial 108 during testing.

## 5 System architecture

The ESP32-WROOM-32 is the chair's brain: it reads the biosignal and sensors, decides the move, and drives the Avishkaar motors. The app talks only to the ESP32.

```
Electrodes → BioAmp EXG Pill (3.3 V) → ESP32-WROOM-32 ADC
                                       ├─ filter + detect glance / blink / clench → command
                                       ├─ BLE ⇔ companion app (prototype UI)
                                       ├─ MAX30102, MPU6050, HC-SR04, water sensor
                                       └─ motor driver (or Maker Board 2.0) → Avishkaar motors
```

| Signal | Action |
|---|---|
| Double blink | Start / stop |
| Look left / right (EOG) | Turn left / right |
| Jaw clench (EMG) | Emergency stop |
| Obstacle < 20 cm, tilt, or water detected | Forced stop (overrides all) |

**Power:** separate batteries for the motors and for the ESP32 + BioAmp, sharing only ground, so motor noise does not swamp the signal.

## 6 Bill of materials and budget

New spend is about ₹6,700–7,500; the chassis and motors come from kit already owned. Prices other than the neuroscience kit are approximate.

| Part | Qty | Approx. ₹ | Buy when |
|---|---|---|---|
| DIY Neuroscience Kit – Basic (without UNO) | 1 | 5,999 | Stage 1 |
| ESP32-WROOM-32 DevKit, pins soldered | 1 | 350–500 | Stage 1 |
| Jumper wires M–M, M–F, F–F (40 each) | 1 set | 150–250 | Stage 1 |
| Breadboard + USB data cable | 1 each | 150–250 | Stage 1 |
| Motor driver TB6612 / L298N | 1 | ~150 | Stage 2 |
| MAX30102, MPU6050, 2× HC-SR04, water sensor, buzzer | 1 set | 600–900 | Stage 3 |
| Avishkaar Pro kit, Maker Board 2.0 | — | Owned | — |

## 7 Build plan and timeline

Work starts around 19–25 Oct; the 30 Oct bonus is a stretch, and the 30 Nov Stage 1 deadline is the real target.

- ☐ Stage 0 (free): learn ESP32 basics on Wokwi; build the chassis shape; draft the PPT story
- ☐ Stage 1: blink, glance and clench detected by the ESP32 at least 8 times in 10
- ☐ Stage 2: ESP32 drives the Avishkaar motors from a simple app joystick
- ☐ Stage 3: biosignal drives the chair with motors running; obstacle stop works
- ☐ Stage 4: health sensors, fall SOS, emergency response, flood mode
- ☐ Stage 5: prototype app UI, demo video, PPT, submit before 30 Nov
- ☐ Finale: Internationals in Delhi, 15–16 Jan 2027 (if selected)

## 8 Risks and open questions

- Motor noise corrupting the biosignal → separate batteries, short twisted electrode wires.
- False blinks → double-blink rule, per-user calibration, filtering.

- ESP32 ADC noise → steady 250–500 Hz sampling and a moving-average filter.
- **Open:** does the Makeathon allow non-Avishkaar parts? Ask support@avishkaar.cc.
- **Open:** can the Maker Board 2.0 take commands from the ESP32, or do we use our own motor driver?

## 9 Sources


- DIY Neuroscience Kit – Basic: store.upsidedownlabs.tech/product/diy-neuroscience-kit-basic
- ESP32 + BioAmp EXG Pill: crowdsupply.com/upside-down-labs/bioamp-exg-pill
- BioAmp EXG Pill examples: github.com/ChinmayLonkar/BioAmp-EXG-Pill
- Smart Wheel Chair: community.avishkaar.cc/project/smart-wheel-chair
- Smart Assistive System for Paralysis People: ijraset.com
- Avishkaar League 2026: community.avishkaar.cc/avishkaar-league-2026